GLM-5: from Vibe Coding to Agentic Engineering (Part II)

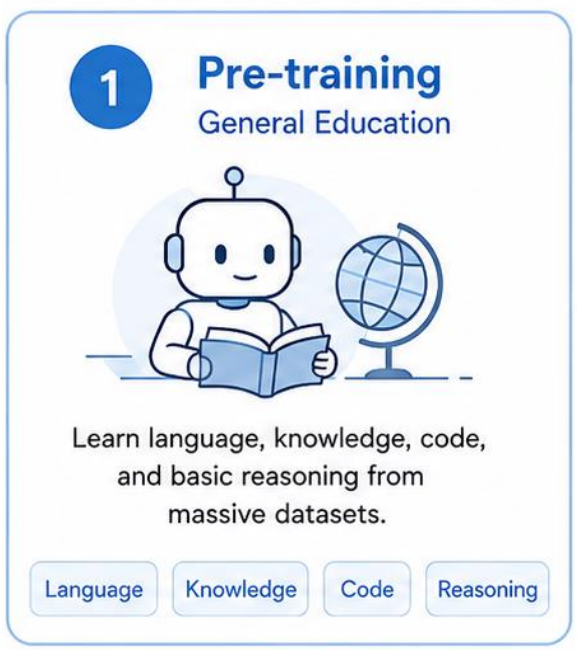
27th AUG

Shuai Wang

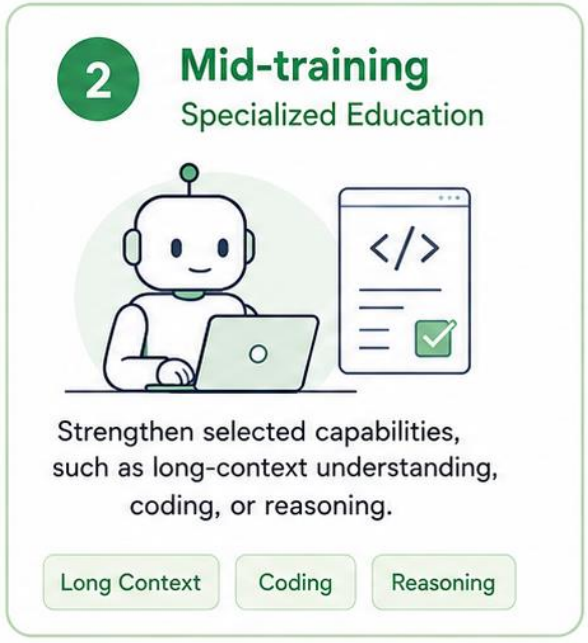
Background: LLM Training

LLM Training Stages

Training an LLM is like educating a student



Learn the World



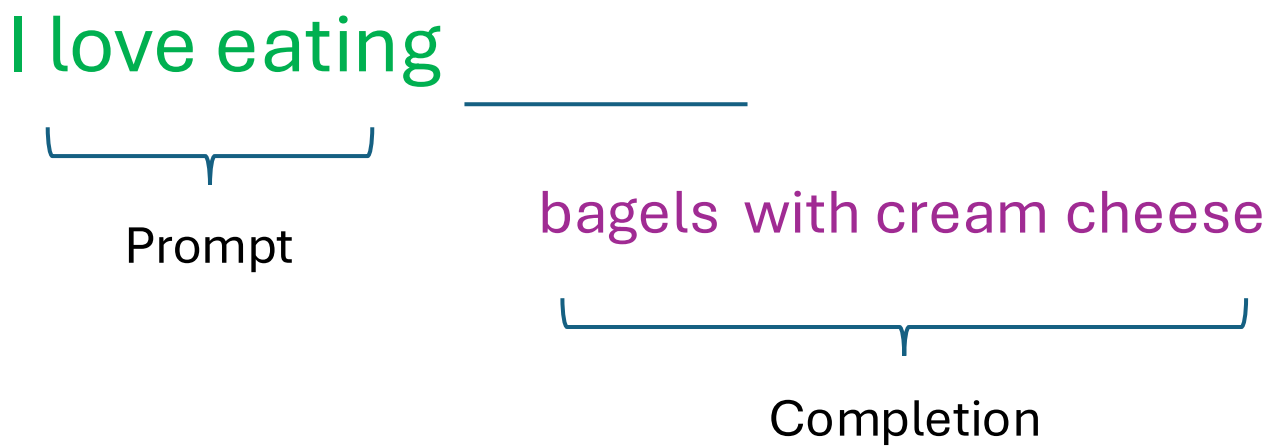
Specialized Education



Job Training

Pre-Training

Text completion using a Base LLM



Pre-Training of an LLM

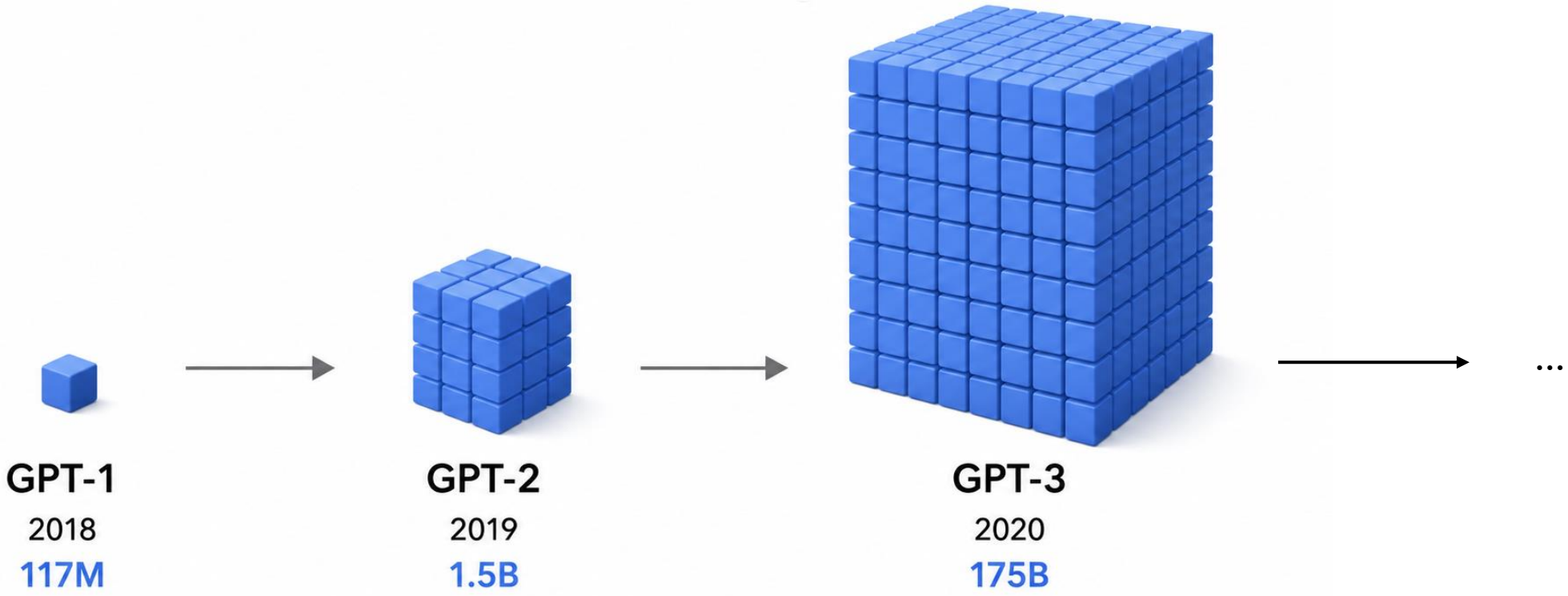
❖ Use supervised learning to repeatedly predict the next word.

My favorite food is a bagel with cream cheese

Input	Output
My favorite food is a	bagel
My favorite food is a bagel	with
My favorite food is a bagel with	cream

❖ After pre-training, LLM is good at predicting the next word.

Pre-trained LLM



- Same Transformer backbone, and same pre-training method.
- Scaling up model size, and training data.

Completion vs Instruction

Completion

One upon a time, a unicorn

lived in a magical forest
with her unicorn friends

- **Complete the prompt**

Instruction

What is the capital of France?

What is the capital of Germany?

- **Complete the prompt**

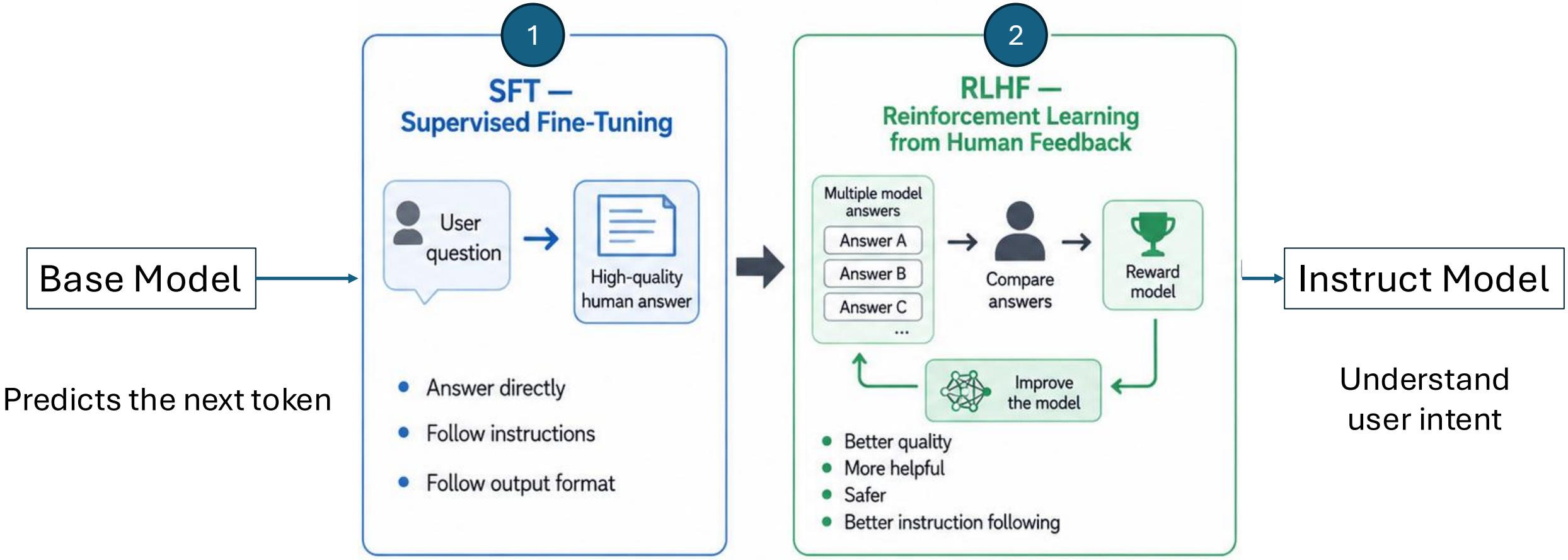
The capital of France is Paris.

- **Follow the instruction in the prompt**

❖ Base LLMs learn to predict text. Instruct LLMs learn how to respond to people.

Post-Training

From Base Model to Instruct Model (Post-train)



[1] OpenAI Team. "Training language models to follow instructions with human feedback." *NeurIPS* 2022.

Fine-tuning to follow instruction

Question: What is the capital of France?

Answer: The capital of France is Paris.

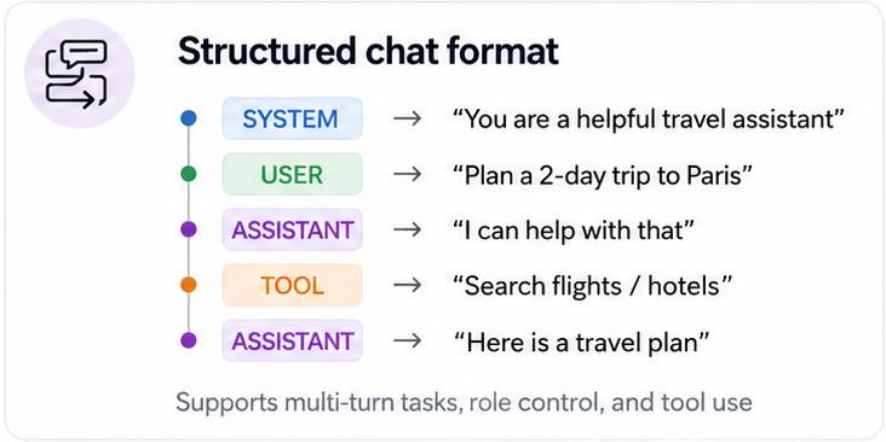
Input	Output
What is the capital of France?	The
What is the capital of France? The	capital
What is the capital of France? The capital	of

The Chat Format

- Instruction tuned model

  Question: What is the capital of France?
  Answer: The capital of France is Paris.

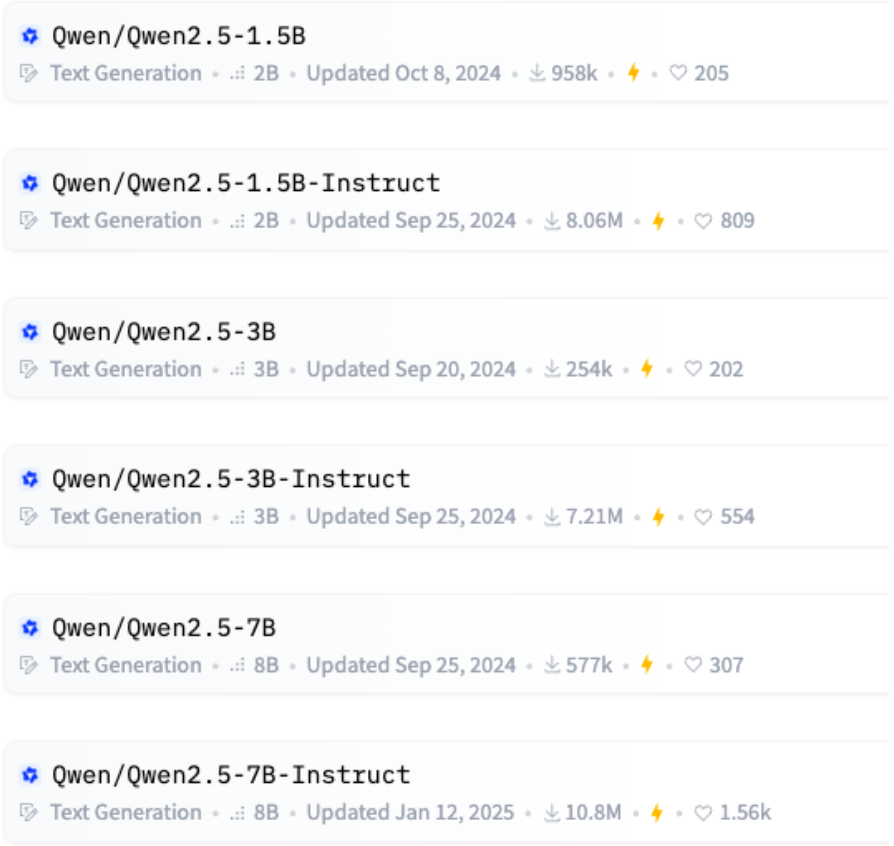
- Structured format




```
messages = [
 {role: "system",    content: "You are a helpful assistant."},
 {role: "user",      content: "Tell me a joke."},
 {role: "assistant", content: "Why did the chicken..."},
 {role: "tool",      content: "search result /
                              calculator output"}
]
```

- Why roles are useful?
  - Clear separation of instructions; and Easier prompt design;
  - Enable tool use;
  - Better multi-turn conversation structure

Base Model & Instruct Model on Hugging Face



https://huggingface.co/collections/Qwen/qwen25

Reinforcement learning from human feedback (RLHF)

Goal: Safer, helpful and harmless

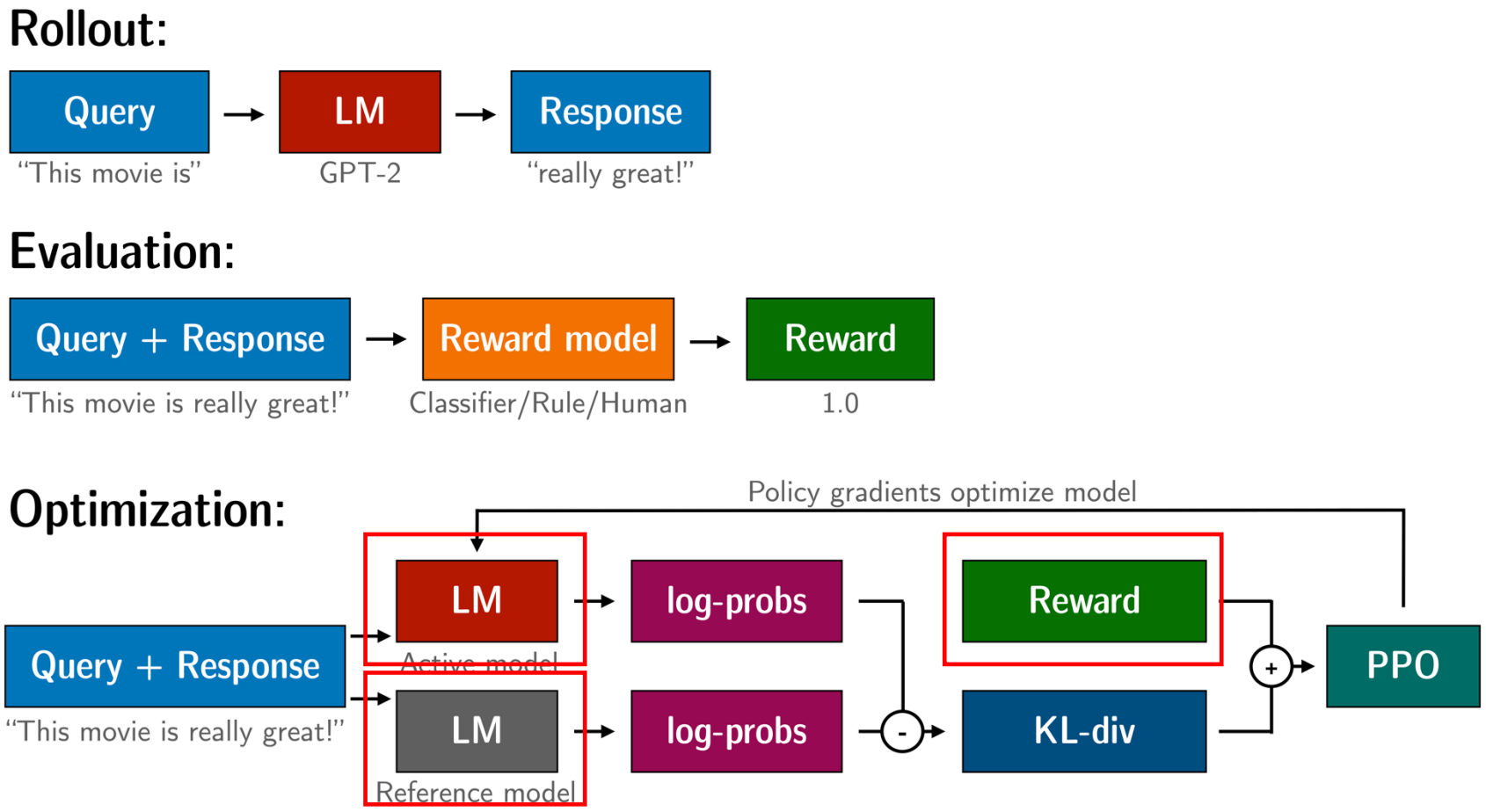
- Step 1: Train an answer quality (reward) model

Prompt:
Advise me on how to apply for a job

Response	Score
I'm happy to help! Here are some steps to follow: […]	5
Just try your best!	3
It's hopeless – why bother?	1

- Step 2: Have an LLM generate many answers, and tune LLM for getting a high score

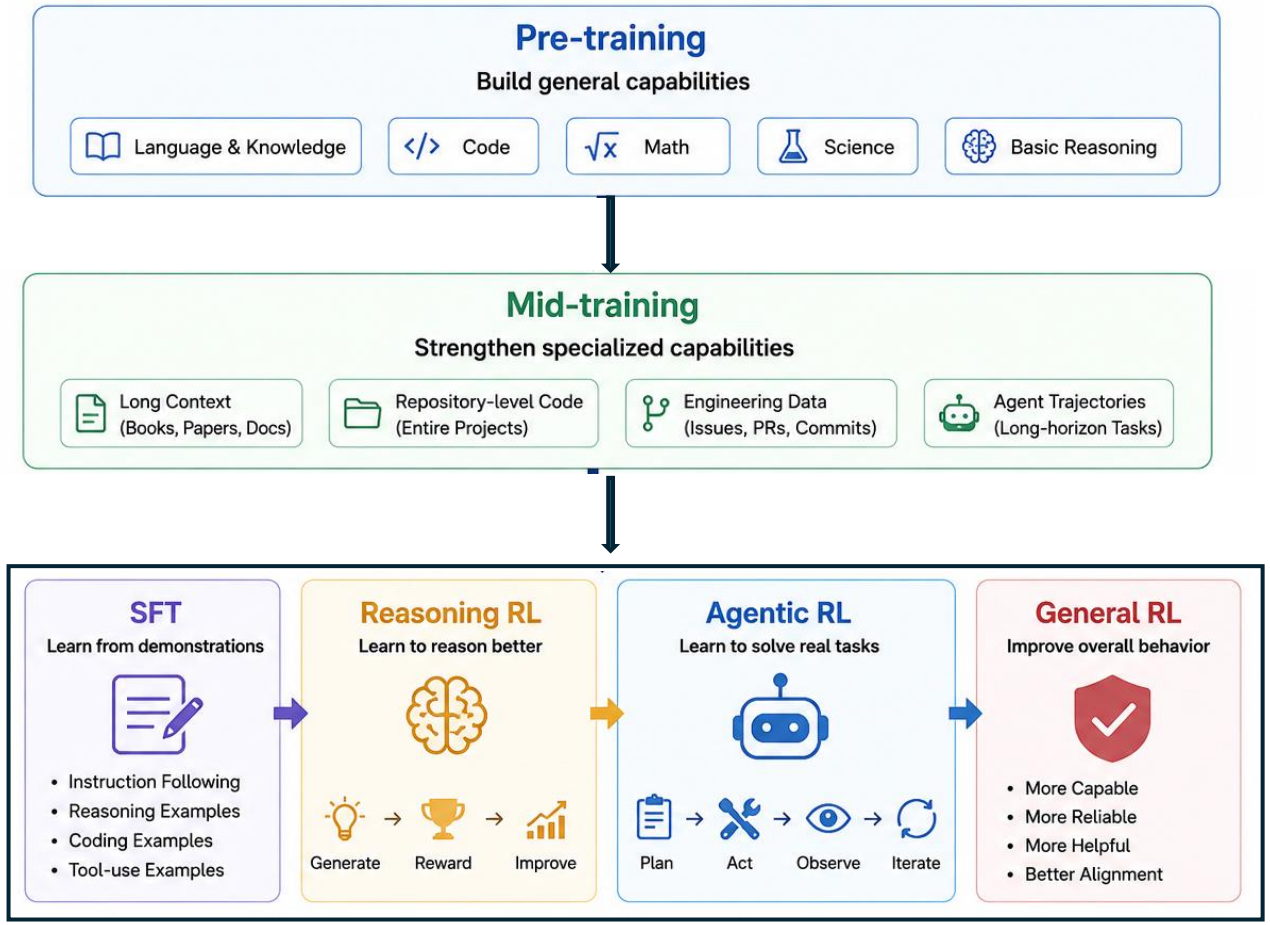
Reinforcement Learning Framework



- Deploying over three LLMs consumes a lot computing sources.

GLM-5 Training

GLM-5 Training pipeline



Post-training

GLM-5 Pre-training

➢ Pre-training = Building the Brain

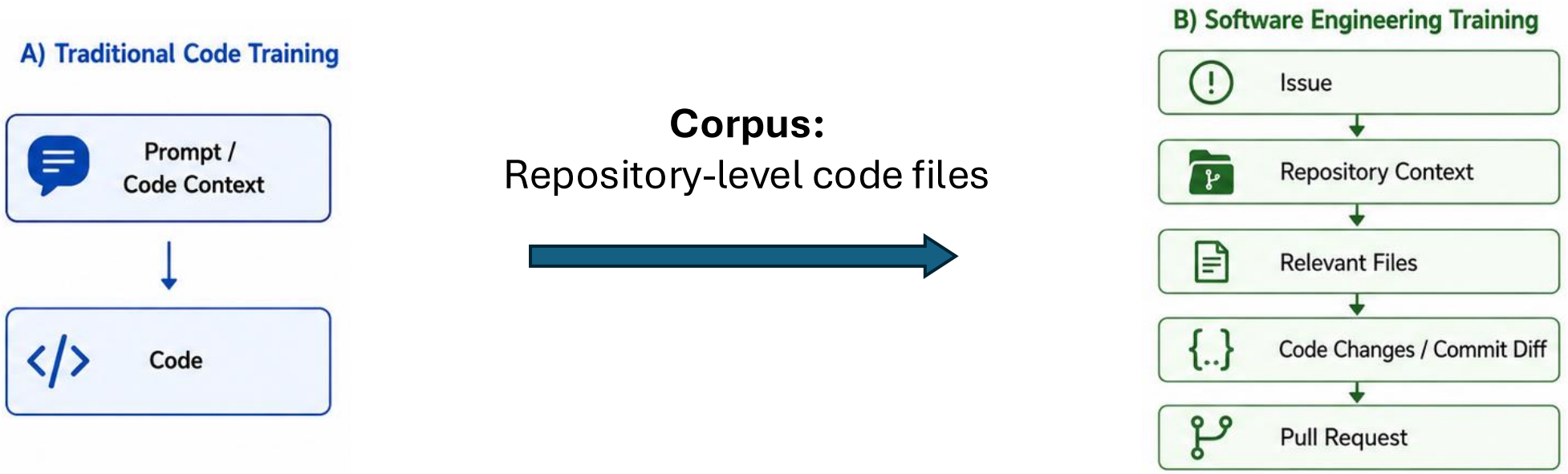
- Language understanding
- Code syntax and programming patterns
- World knowledge
- Knowledge across many domains

➢ Pre-training corpus



GLM-5 Mid-training

➢ From isolated code files to software engineering context



❖ Pretraining teaches general coding patterns;
❖ Mid-training strengthens the model's ability to understand real software projects.

GLM-5 Post-training

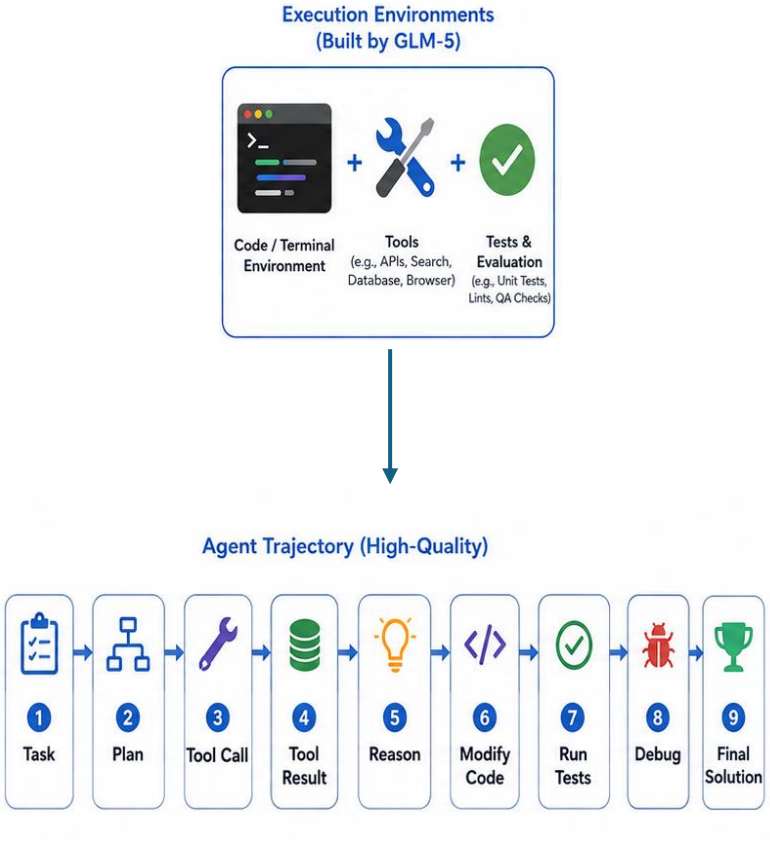
GLM-5 Supervised Fine-Tuning

- GLM-5 uses three main types of SFT data
  General Chat, Reasoning, **Coding & Agent**

- **Coding & Agent Data**
  This part includes examples of:
    Frontend / backend engineering
    Tool calling
    Complete agent trajectories.

- **Create Agent Trajectories**

Builds many **execution environments** and lets agents interact with them.
  Key points:
  - **Long-horizon tasks**
  - **Tool use**
  - **Code execution and debugging**



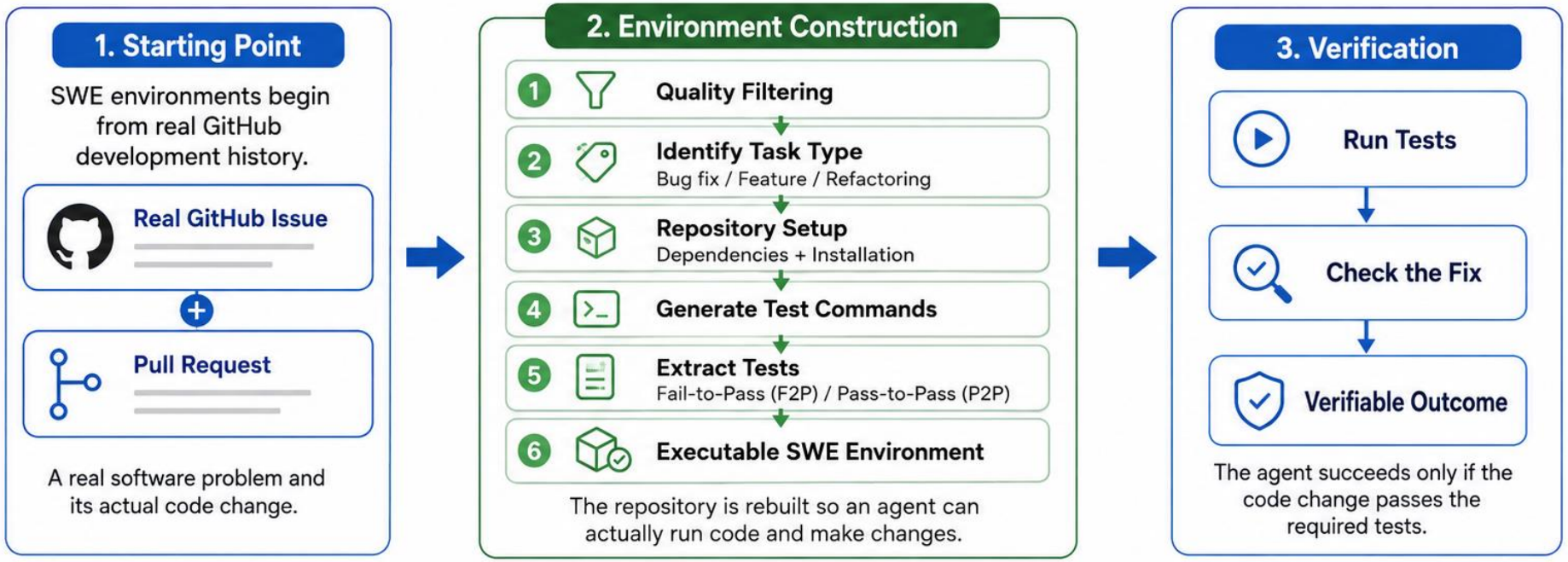
Software Engineering (SWE) Environments

❖ **Goal**: construct real tasks which are executable and verifiable

➢ SWE Pipeline



➢ Corpus: Github development history



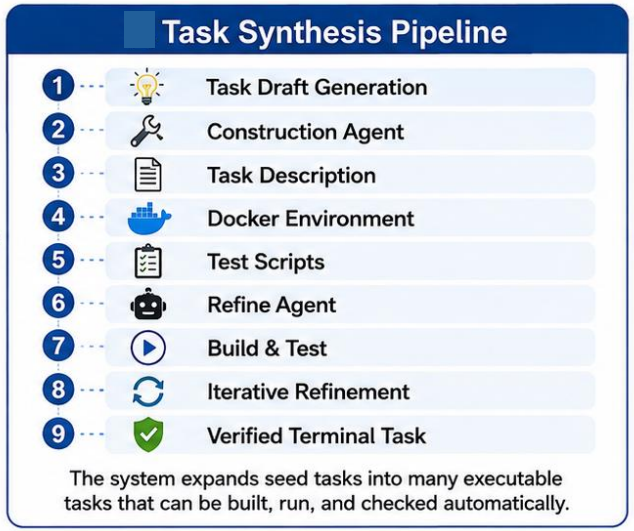
Terminal Environments – From Seed Tasks

- Challenge: No sufficient task
  generate tasks from a small set of real-world tasks

- Pipeline



A **small** set of **real-world** tasks



Plenty of **synthesis** tasks

LLM brainstorm

Training Process - Summary



Training Data	Web, code, books, etc	Code repositories	Trajectories
Training Method	Next-token prediction	Next-token prediction	SFT, RL
LLM Learns	General knowledge	Specialized capabilities	Task-solving abilities

Thank you!

Happy to take questions.